

Charan Lingamgunta

STAFF SOFTWARE ENGINEER

DISTRIBUTED SYSTEMS | DEVELOPER PLATFORMS | AI INFRASTRUCTURE

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PROFESSIONAL SUMMARY

Staff Software Engineer with 15 years of experience leading distributed systems, developer platforms, and AI infrastructure in regulated enterprise environments. Delivered platform capabilities supporting 10B+ daily transactions, 200+ engineers, 1,000+ APIs, and 20+ production models across healthcare, finance, and government clients. Track record of driving architecture, technical standards, and cross-team execution for secure, high-scale platforms.

CORE SKILLS

Platform & Distributed Systems: Distributed Systems, Event-Driven Architecture, Pub/Sub & Messaging (JMS, AMQP), API Gateways, Developer Platforms, Multi-Tenant Systems, High-Availability Design

AI & ML Platforms: Agent Harness, Agent Platforms, RAG, LLM Fine-Tuning, Model Evaluation, MCP, DSPy, MLflow, Guardrails, Hugging Face, PyTorch, CUDA, Ray, vLLM, PEFT, LoRA

Languages: Java, Python, TypeScript, JavaScript, C#

Cloud & Infrastructure: AWS (EC2, S3, RDS, ECS, Lambda, CloudWatch), Docker, Terraform, OpenSearch, Airflow, Jenkins, GitHub Actions, SkyPilot

Frameworks & Developer Tooling: Backstage, React, Node.js, FastAPI, Django

Security & Compliance: OAuth 2.0, SAML, LDAP, HSM, HIPAA, HITRUST, PCI-DSS, SOC 2, FIPS 140-2

PROFESSIONAL EXPERIENCE

Staff Software Engineer | Forum Systems | Needham, MA

Jun 2014 - Present

Progressed from core API gateway engineering into staff-level ownership across developer platform, agent execution, and LLM platform initiatives while shaping shared architecture, security standards, architecture reviews, and ADR practices across teams.

Concurrent Platform Ownership

DevX, Developer Experience Platform

Jan 2022 - Present

- Led architecture for an enterprise developer platform serving Fortune 500 healthcare and financial services clients, with 200+ engineers and 1,000+ APIs across client organizations, unifying API discovery, deployment visibility, dependency mapping, and operational metadata.
- Designed an OpenAPI-to-tool pipeline that converted 1,000+ API specifications into 2,500+ governed tools, alongside publication workflows for custom non-API tools built from approved Python libraries, and self-serve authoring workflows that let analysts compose pre-tested tool pipelines from engineer-published templates.
- Delivered multi-tenant capabilities for regulated enterprise deployments, including client-specific plugins, flexible Okta/GitHub/local identity integration, and entity-level permissions for APIs, resources, and AI features.
- Improved engineering leverage by establishing architecture reviews and ADR practices and mentoring 5 engineers, including 2 promoted to senior roles.

AI SecureExec, Agent and Tool Execution Platform

Feb 2023 - Present

- Built an enterprise agent harness (DSPy-based) handling 10,000+ daily invocations across the six harness components (orchestration, governance, state, memory, observability, and evals), enabling analyst and data science teams at enterprise clients to run DevX-approved tools through ReAct and CodeAct workflows in regulated environments.
- Engineered an isolated Python execution plane with containerized sandboxing, runtime timeouts, restricted

network and filesystem access, and safeguards against malicious code, infinite loops, and data exfiltration.

- Published an MCP-compatible tool registry for approved API-derived and custom-authored tools, giving teams a governed path to reuse enterprise capabilities in agent workflows.
- Designed multiple RAG pipelines (classic one-shot and agentic ReAct loops) over OpenSearch hybrid (BM25 + kNN) retrieval, ingesting 50,000+ documents monthly via PDF/Office chunking, embedding generation, scheduled API ingestion, and event-driven updates.
- Implemented a multi-provider LLM gateway via LiteLLM, routing per-tenant traffic across self-hosted and fine-tuned models alongside Bedrock-hosted OpenAI and Anthropic models, with per-tenant access policies, fallback chains, and per-call cost emission and budget enforcement.
- Implemented content-moderation policies for PII detection, redaction, and safety guardrails, and added MLflow/OpenSearch observability for agent execution, tool usage, and policy decisions.

DevX Fine-Tuning, LLM Fine-Tuning and Deployment Platform

Jan 2023 - Jun 2025

- Developed an LLM fine-tuning platform supporting 20+ production models deployed across enterprise client environments, giving teams a consistent workflow to onboard Hugging Face models, configure prompts and templates, launch jobs, and compare runs in MLflow.
- Designed reusable dataset pipelines for Excel and JSON training corpora, with versioned datasets, repeatable preprocessing, and reproducible train/validation/test splits.
- Supported parameter-efficient fine-tuning and checkpoint selection through held-out test sets, MLflow run comparisons, and LLM-as-judge plus human review before release.
- Operationalized GPU-backed vLLM serving across AWS and GCP, using SkyPilot spot orchestration and Ray-based distributed training to lower training cost and OpenSearch to monitor training and production behavior.

Forum Sentry, Enterprise API Gateway and Builder

Jun 2014 - Dec 2021

- Architected a 4-stage bidirectional streaming pipeline with zero-copy passthrough, automatic retry/failover, and 7 timing checkpoints for latency attribution across 10B+ daily transactions over HTTP, SOAP, JMS, AMQP, SFTP, and S3 protocols.
- Built event-driven integration capabilities across enterprise messaging systems (JMS, AMQP, IBM MQ, Solace, TIBCO, ActiveMQ), including pub/sub patterns, retry/failover policies, and async processing pipelines for mission-critical financial and government workloads.
- Designed a no-code workflow engine with 60+ pre-built processors for security, transformation, routing, validation, and integration, enabling complete API implementations that connect to core systems through custom-built adapters, reducing client onboarding from weeks to days.
- Engineered active-passive failover and 9 load-balancing strategies, supporting sub-minute recovery and 99.99% uptime targets for mission-critical deployments.
- Implemented the authentication subsystem supporting 11 enterprise identity methods, with multi-tier caching and connection pooling that eliminated 90%+ of directory roundtrips.
- Engineered multi-tier caching across memory, disk, and Redis with RFC 7234 compliance, request coalescing, and no-code cache policy configuration, reducing backend load by 40-70%.

Software Developer Intern | Mimecast | Watertown, MA

Apr 2013 - Aug 2013

- Contributed to an enterprise email encryption application in C# and WPF and added PowerShell-based test automation.

Associate Software Engineer | AppLabs (CSC) | Hyderabad, India

Dec 2009 - Jun 2012

- Developed reporting and notification modules for an e-learning platform serving millions of users.

EDUCATION

Master of Science in Computer Information Systems, Northeastern University

Boston, MA

Bachelor of Engineering in Computer Science, JNT University

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